

## Toxic Threat Zone

ALOHA® 5.4.7



Time: July 21, 2025 1751 hours PDT (using computer's clock)

Chemical Name: AMMONIA

Wind: 1.55 meters/second from 271° true at 10 meters

### THREAT ZONE:

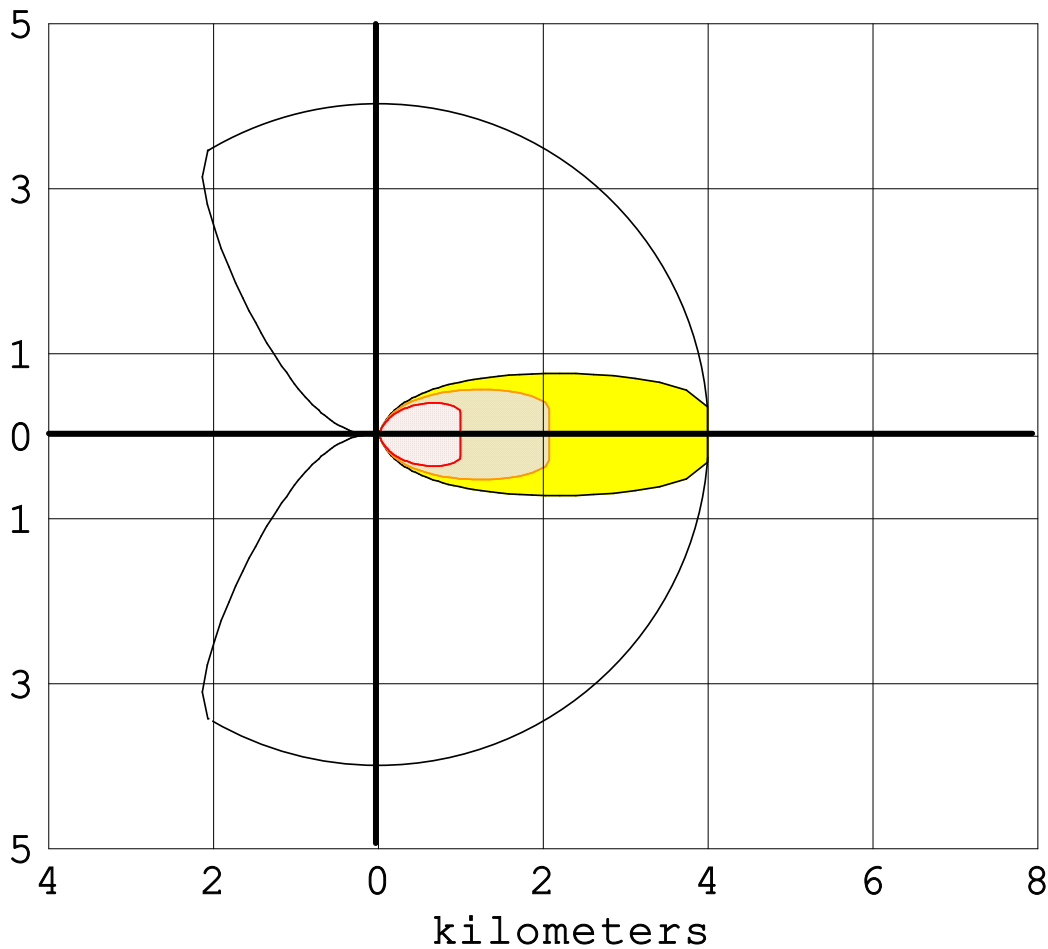
Model Run: Heavy Gas

Red : 1.0 kilometers --- (1100 ppm = AEGL-3 [60 min])

Orange: 2.1 kilometers --- (160 ppm = AEGL-2 [60 min])

Yellow: 4.0 kilometers --- (30 ppm = AEGL-1 [60 min])

kilometers



-  greater than 1100 ppm (AEGL-3 [60 min])
-  greater than 160 ppm (AEGL-2 [60 min])
-  greater than 30 ppm (AEGL-1 [60 min])
-  wind direction confidence lines

## Text Summary

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### SITE DATA:

Location: TULARE, CALIFORNIA, USA, CALIFORNIA  
Building Air Exchanges Per Hour: 0.28 (unsheltered single storied)  
Time: July 21, 2025 1751 hours PDT (using computer's clock)

### CHEMICAL DATA:

Chemical Name: AMMONIA  
CAS Number: 7664-41-7 Molecular Weight: 17.03 g/mol  
AEGL-1 (60 min): 30 ppm AEGL-2 (60 min): 160 ppm AEGL-3 (60 min): 1100 ppm  
IDLH: 300 ppm LEL: 150000 ppm UEL: 280000 ppm  
Ambient Boiling Point: -33.6° C  
Vapor Pressure at Ambient Temperature: greater than 1 atm  
Ambient Saturation Concentration: 1,000,000 ppm or 100.0%

### ATMOSPHERIC DATA: (MANUAL INPUT OF DATA)

Wind: 1.55 meters/second from 271° true at 10 meters  
Ground Roughness: open country Cloud Cover: 4 tenths  
Air Temperature: 18.54° C  
Stability Class: D (user override)  
No Inversion Height Relative Humidity: 54%

### SOURCE STRENGTH:

Leak from hole in vertical cylindrical tank  
Flammable chemical escaping from tank (not burning)  
Tank Diameter: 2 meters Tank Length: 3 meters  
Tank Volume: 9.42 cubic meters  
Tank contains liquid Internal Temperature: 18.54° C  
Chemical Mass in Tank: 4,908 kilograms  
Tank is 85% full  
Circular Opening Diameter: 5 centimeters  
Opening is 2.55 meters from tank bottom  
Note: RAILCAR predicts a stationary cloud or 'mist pool' will form.  
Model Run: traditional ALOHA tank  
Release Duration: 23 minutes  
Max Average Sustained Release Rate: 937 kilograms/min  
(averaged over a minute or more)  
Total Amount Released: 1,571 kilograms  
Note: The chemical escaped as a mixture of gas and aerosol (two phase flow).

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